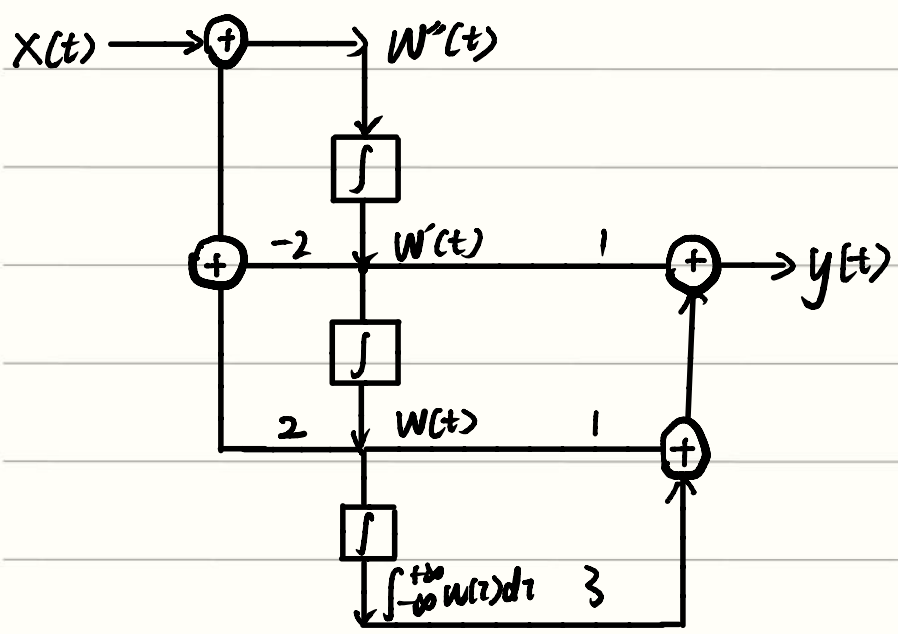
2-25.(3):

令$W(t)$为系统的中间变量

则 $\frac{d^2w(t)}{dt^2}+2\frac{dw(t)}{dt}-2w(t)=x(t)$

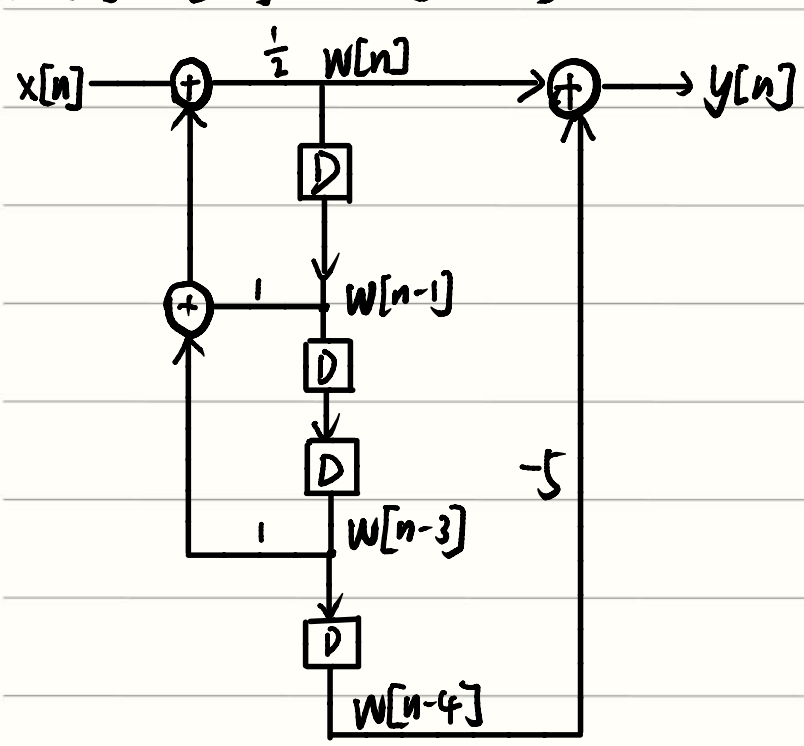
$w''(t)=x(t)-2w'(t)+2w(t)$



2-26.(1):

令中间变量为$w[n]$，则：

$2w[n]-w[n-1]+w[n-3]=x[n]$



补充题

(1): $y(t)=x(t)*h(t)=h(t)*x(t)$

$=\int_{-\infty}^{+\infty}h(\tau)x(t-\tau)d\tau$

$=\int_{-\infty}^{+\infty}(2e^{-\tau}-3e^{-4\tau})u(\tau)e^{5(t-\tau)}d\tau$

$=e^{5t}\int_{-\infty}^{+\infty}(2e^{-6\tau}-3e^{-9\tau})u(\tau)d\tau$

$=e^{5t}\int_{0}^{+\infty}(2e^{-6\tau}-3e^{-9\tau})d\tau$

$=e^{5t}\cdot\left[\left(-\frac{2}{6}e^{-6\tau}\right)\Big|_0^{+\infty}-\left(-\frac{3}{9}e^{-9\tau}\right)\Big|_0^{+\infty}\right]$

$=e^{5t}\left(-\frac{1}{3}-\left(-\frac{1}{3}\right)\right)$

$=0$

(2): $\int_{-\infty}^{+\infty}\left|(2e^{-t}-3e^{-4t})u(t)\right|dt$

原式$<2\int_{-\infty}^{+\infty}\left|e^{-2t}\cdot u(t)\right|dt+3\int_{-\infty}^{+\infty}\left|e^{-4t}u(t)\right|dt$

$=2\int_0^{+\infty}e^{-2t}dt+3\int_0^{+\infty}e^{-4t}dt$

$=2\cdot\frac{1}{-2}e^{-2t}\Big|_0^{+\infty}+3\cdot\frac{1}{-4}e^{-4t}\Big|_0^{+\infty}$

$=-1(0-1)-\frac{3}{4}(0-1)$

$=\frac{7}{4}<+\infty$

∴是稳定系统